Linglong Meng

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EDUCATION

University of Minnesota, Twin Cities **Sep. 2024 - May 2026**

Master of Science in Computer Science *GPA: 3.49 | Major: 3.73*

- Master Project: Extending Sterling with OCaml Code Generation for Extensible Language Specifications
- Relevant Coursework: Theory of Programming Languages, Modern Compiler Implementation in Standard ML, Software Foundations (Rocq), Elementary Mathematical Logic

University of Minnesota, Twin Cities **Sep. 2020 - May 2024**

B.S. in Computer Science; Minors: Mathematics, Asian & Middle Eastern Studies *GPA: 3.43 | Major: 3.53*

Honors: Dean's List (Fall 2022 - Spring 2024)

RESEARCH EXPERIENCE

UMN-Twin Cities – Minnesota Extensible Language Tools Group Sep. 2023 – Present

Research Assistant | Advisor: Dr. Eric Van Wyk

- Built a VS Code extension to enhance debugging support in Silver, an attribute grammar language.
- Extended the Silver-based system Sterling to generate OCaml interpreters from structured operational semantics (syntax, judgments, inference rules), previously supporting only Prolog/λProlog targets.
- Presented a poster: Compiling Structured Operational Semantics to Executable OCaml Code, Midwest Programming Languages Summit (MWPLS), 2025.

Shanghai Jiao Tong University – Programming Languages and Verification May 2024 – Nov. 2024

Research Internship, Hybrid | Advisor: Dr. Yuting Wang

- Developed a formally verified translation path from Rust's intermediate representation (IR) to Clight, the C subset of ***CompCert***, a verified optimizing compiler mechanized in the Rocq (Coq) proof assistant.
- Proved semantic preservation via small-step simulation within CompCertO's framework of open LTS and Kripke-style world semantics, ensuring compositional correctness and modular reasoning under verified compilation.
- Contributed to "End-to-end Compositional Verification of Program Safety through Verified and Verifying Compilation." arXiv preprint, arXiv:2510.10015, 2025.

RELEVANT PROJECTS & COMPETITIONS

sdk_specify (Personal Dafny Project) Aug. 2022 – Sep. 2022

GitHub | Advised by Ryan Emery, AWS

- Implemented and formally verified AWS Key Management Service (KMS) Amazon Resource Name (ARN) parsing and validation logic in Dafny, aligned with the AWS Encryption SDK specification.
- Refined proof structure, which later helped form part of an internal AWS tutorial on verified SDK components.

***1st place*, MinneHack 2023** Jan. 28 – 29, 2023

Hackathon | *bill-nom project link* | *Run by UMN ACM Student Chapter*

- Bill Nom digests large legal documents into a compact summary using NLP via a URL from the MN Revisor's Office, scrapes it using Beautiful Soup 4 (BS4), with a T5 NLP model.
- Our 4-person team won 1st place among 43 projects of a total 165 participants.

TEACHING & LEADERSHIP EXPERIENCE

iD Tech, Remote May 2023 - Aug. 2023

Online Instructor

- Instructed around 30 students aged 10-15 on Java and Python via Zoom.
- Successfully led students to code their first simple game "Tic-Tac-Toe" in a 5-lesson curriculum.

Japanese Social Justice Organization Jan. 2025 - Present

President

- Organized 7 events (4 academic, 3 social) averaging 25 participants, maximum 55.
- Managed over $3,000 organizational budget.
- Attracted 90 new Instagram followers, growing total following to 480.

TECHNICAL SKILLS

Programming: Rocq (Coq), OCaml, Rust, Standard ML, Silver, Dafny, C, Python, MATLAB

Languages: English (fluent), Chinese (native)